

## Machine Project

### Major Details

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| <b>Groupings:</b>             | At most 3 members in a group       |
| <b>Deadline:</b>              | March 05, 2026 (Thursday) 6:00 PM  |
| <b>Demo Schedule:</b>         | March 10-12, 2026                  |
| <b>Percentage:</b>            | 35%                                |
| <b>Submission guidelines:</b> | Submit the zip file to Blackboard  |
| <b>Filename format:</b>       | AI3-Project-<Section>-Group<#>.zip |

### Deliverables

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Zip file containing:

- Jupyter Notebook file (.ipynb)
- Python source code files (.py)
- Dataset (or a link and description of the dataset used)
- Web deployment code (Streamlit or Gradio scripts) and Full System Demo (Video)
- Presentation File (PPT / Canva).
- Documentation in IEEE paper format (.doc/.docx)
- A copy of the project folder generated during training (results, logs, model weights, evaluation metrics, etc.)
- Evidence of model testing on an unseen test set, clearly separated from training/validation artifacts

### Specifications

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This project focuses on the design and implementation of a Large Language Model (LLM) chatbot. Each group is required to select an appropriate knowledge source (e.g., PDF documents, research papers, manuals, policies, websites, or instructor-approved corpora) that will serve as the chatbot's information base. The dataset may be closed-corpus or hybrid, subject to topic approval.

Once the knowledge sources have been selected, the group must preprocess and index the data, including document cleaning, chunking, embedding generation, and storage in a suitable vector database. The group must then integrate an LLM with the retrieval component to enable context-aware question answering, ensuring that responses are grounded in the retrieved documents.

The system must be evaluated using appropriate qualitative and/or quantitative metrics (e.g., relevance, response accuracy, faithfulness, latency, and user evaluation). The final chatbot must be deployed through a web-based interface developed using Streamlit or Gradio, allowing users to interactively query the system.

All project artifacts: including the indexed documents, embedding store, configuration files, logs, and deployment code, must be preserved and included in the project submission.

Documentation must strictly follow the IEEE paper format, including abstract, introduction, related work (if applicable), methodology, results, discussion, conclusion, and references.

As part of the project requirements, every group must schedule at least one consultation with the instructor during the term. **This consultation should be completed no later than one week before the deadline and should cover the group's partial work.** Completion of this requirement forms part of the grading scheme for the project.

## **Outline for the Notebook**

### **Section 1. Introduction to the Problem/Task**

Introduce the chatbot problem being addressed. Clearly describe the purpose of the LLM chatbot, the domain or use case (e.g., academic documents, policies, manuals, FAQs, research papers), and the real-world significance of building a retrieval-grounded conversational system.

### **Section 2. Dataset Description**

In this section of the notebook, you must fulfill the following:

- Provide a brief description of the knowledge sources used (e.g., PDFs, web pages, text files, databases).
- State the source of the documents and how they were collected.
- Explain the dataset structure (number of documents, file types, size, domains).
- Discuss any preprocessing steps applied (cleaning, chunking strategy, token limits, metadata tagging, document filtering).
- Describe the embedding process (model used, chunk size, overlap).

### **Section 3. Requirements**

List all Python libraries, frameworks, and tools used, such as:

- LLM frameworks (e.g., LangChain, LlamaIndex)
- Embedding models
- Vector databases (e.g., FAISS, Chroma, Pinecone)
- Backend and UI tools (Streamlit, Gradio)
- Any additional utilities (PDF loaders, web scrapers, etc.)).

### **Section 4. System Architecture**

- Describe the overall architecture (retriever, vector store, LLM, prompt template).
- Explain the pipeline (query → embedding → similarity search → context injection).
- Present prompt design and grounding strategy.
- Include system flow diagrams or pseudocode (if applicable)

## Section 5. System Evaluation (Unseen Queries)

Explain how the chatbot was evaluated using unseen or held-out queries that were not used during development.

- Describe the evaluation setup (manual testing, benchmark questions, user simulation).
- Report relevant metrics (e.g., response relevance, accuracy, faithfulness, latency).
- Discuss hallucination handling and failure cases.
- Compare retrieved context vs. final generated answers.

## Section 6. Web Deployment

- Develop a **Streamlit** or **Gradio** interface.
- Allow users to input questions or prompts.
- Display retrieved context (optional but encouraged).
- Show chatbot responses in real time.
- Include screenshots of the deployed web application.

## Section 7. Results and Analysis

- Present qualitative results (sample Q&A interactions).
- Present quantitative or structured evaluation results (if applicable).
- Discuss strengths, weaknesses, edge cases, and observed limitations.
- Analyze how retrieval quality affects response quality.

## Section 8. Documentation (IEEE Format)

Final project paper must include:

1. Abstract
2. Introduction
3. Methodology (knowledge base, pipeline, LLM, deployment)
4. Results and Discussion
5. Conclusion and Future Work
6. References (IEEE style)

## Section 9. Insights and conclusions

Summarize what your group learned about building a **LLM chatbot**. Discuss system strengths, limitations (e.g., retrieval errors, hallucinations), and propose areas for future improvement such as better embeddings, reranking, or hybrid retrieval.

## Section 10. References

Cite relevant references that you used in your project. All references must be cited, including:

- **Scholarly Articles**
  - Cite in APA format, and put a description of how you used it for your work.
- **Online references, blogs, articles that helped you come up with your project**
  - Put the website, blog, or article title, link, and how you incorporated it into your work.
- **Artificial Intelligence (AI) Tools**
  - Put the model used (e.g., ChatGPT, Gemini), the complete transcript of your conversations with the model (including your prompts and its responses), and a description of how you used it for your work.

### **Final Project Presentation**

Here are some guidelines regarding the final project presentation:

- Each group is given 45 minutes: 25 minutes to present, and 20 minutes for Q&A.
- Presentations will be done either online or face-to-face.
- Open all the necessary files before your allotted presentation time slot. Do not wait until the presentation itself to load anything.
- Presentation must include:
  - Live demo of the Streamlit/Gradio app.
  - Jupyter Notebook and the IEEE paper.
- All members should be present and should discuss a part in the final project presentation.
- Kindly read the rubrics to check different requirements and expectations on the project presentation.

### **Working With Groupmates**

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For this project, you are encouraged to work in groups of at most 3 members. Make sure that each member of the group has approximately the same amount of contribution for the project. Problems with groupmates must be discussed internally within the group, and if needed, with the lecturer.

### **Use of Artificial Intelligence Tools**

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You are allowed to use AI tools to assist you in the creation of your work, under the following conditions:

1. You must declare the use of such tools following the prescribed format in the *References* section of your submission (see Section 11 under the outline for the Notebook).
2. You must **not** use any code written by AI directly in your submission.
3. You must validate any AI response through your own understanding of the concepts or through your own research.
4. You must be able to articulate the thought processes, rationales, and implementation details of your work, and through this you must be able to show that human agency was maintained even if AI was used in augmenting the process.

Use of AI outside of these parameters is considered academic dishonesty.

## **Deliverables**

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Submit a ZIP file containing all source code and supporting files via Blackboard. The complete chatbot system including: data loaders, preprocessing scripts, embedding generation, vector store integration, retrieval pipeline, LLM interaction logic, prompt templates, and error handling, must be implemented in Python 3.

All core components of the LLM chatbot must be integrated and demonstrated within a Jupyter Notebook. The notebook should clearly present the system design, implementation, and testing of the chatbot in a structured and reproducible manner. This includes data ingestion, document indexing, retrieval behavior, response generation, and evaluation using sample or unseen queries.

Any additional code, configuration files, datasets (or document samples), vector indexes, logs, or deployment-related files used in the project, aside from those directly shown in the notebook must also be included in the submission.

## **Academic Honesty Policy**

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Honesty policy applies. Please take note that you are NOT allowed to borrow and/or copy-and-paste – in full or in part – any existing related program code or solutions from the internet or other sources (such as printed materials like books, or source codes by other people that are not online). You should develop your own codes and solutions from scratch by yourselves.

## **RUBRIC FOR GRADING**

| <b>Criteria</b>                        | <b>Ratings</b>                   | <b>Points</b> | <b>Description</b>                                                                                                                                                                         |
|----------------------------------------|----------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description of Task and Knowledge Base | COMPLETE / INCOMPLETE / NO MARKS | 5             | Clear overview of the chatbot's purpose, domain, and use case; objectives and rationale for using a LLM-based approach are well defined.                                                   |
| Knowledge Base / Dataset Description   | COMPLETE / INCOMPLETE / NO MARKS | 5             | Source, document types, structure, size, and preprocessing steps (cleaning, chunking, embeddings) are clearly explained.                                                                   |
| Model Training                         | COMPLETE / INCOMPLETE / NO MARKS | 10            | Overall RAG pipeline (retriever, vector store, LLM, prompt design) is appropriate, well designed, and clearly documented.                                                                  |
| Correctness of Training                | COMPLETE / INCOMPLETE / NO MARKS | 10            | Evidence of correct retrieval behavior (relevant context selection) and grounded response generation; issues such as hallucination or irrelevant retrieval are acknowledged and addressed. |
| Model Testing (Unseen Data)            | COMPLETE / INCOMPLETE / NO MARKS | 5             | Uses unseen or held-out queries; reports qualitative and/or quantitative evaluation; discusses generalization, faithfulness, and leakage avoidance.                                        |
| Web Deployment                         | COMPLETE / INCOMPLETE / NO MARKS | 10            | Functional Streamlit/Gradio chatbot interface; accepts user queries and returns responses; retrieval context display is a plus.                                                            |
| Results and Analysis                   | COMPLETE / INCOMPLETE / NO MARKS | 10            | Presents representative chatbot interactions, evaluation results, and logs; interprets strengths, limitations, and failure cases.                                                          |
| Error Handling & Robustness (App)      | COMPLETE / INCOMPLETE / NO MARKS | 5             | App handles invalid inputs, empty queries, and retrieval failures gracefully; reasonable latency and user feedback are provided.                                                           |
| Documentation (IEEE Format)            | COMPLETE / INCOMPLETE / NO MARKS | 15            | Paper strictly follows IEEE format with all required sections and proper citations.                                                                                                        |

| Criteria                      | Ratings                                | Points | Description                                                                                                                             |
|-------------------------------|----------------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Insights and Conclusions      | COMPLETE /<br>INCOMPLETE / NO<br>MARKS | 5      | Thoughtful reflection on system strengths, limitations, and future improvements (e.g., better embeddings, reranking, hybrid retrieval). |
| Notebook & Code Documentation | COMPLETE /<br>INCOMPLETE / NO<br>MARKS | 5      | Clear, reproducible Notebook and well-documented code with instructions for running the chatbot.                                        |
| Consultation                  | COMPLETE / NO<br>MARKS                 | 5      | Group consulted the instructor at least once during project development.                                                                |
| Presentation Manner           | COMPLETE /<br>INCOMPLETE / NO<br>MARKS | 5      | Clear, confident delivery with minimal reliance on notes.                                                                               |
| Presentation Organization     | COMPLETE /<br>INCOMPLETE / NO<br>MARKS | 5      | Logical, engaging presentation flow that is easy to follow.                                                                             |

**Note:** Each member of the group is expected to have a good understanding of the group's submission, even the parts that were not directly delegated to them. Failure to answer the questions during the demo, in a such a way that suggests that one or more group members did not sufficiently understand the work that was delivered, will result in a grade of 0 for those members for the entire project.